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PLGA nanofibers carrying rhFGF1 for the effective treatment of skin wounds in diabetic mice

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Abstract

Diabetic foot ulcers are among the most common complications of diabetes and can lead to delayed wound healing. Fibroblast growth factor (FGF1) is a classic drug for the treatment of skin wounds but has the disadvantages of a short half-life and instability. Poly (lactic-co-glycolic acid) (PLGA) nanofibers are sustained-release biomaterials that have potential for use as therapeutic delivery systems. However, the therapeutic effect of PLGA loaded with recombinant human FGF1 (rhFGF1) on diabetic wound healing is unknown. Therefore, this study aimed to explore the therapeutic effects of PLGA-rhFGF1 in type 2 diabetic (T2D) wound mice. We found that PLGA offers good sustained release, which enhances the stability and bioactivity of rhFGF1. PLGA-rhFGF1 promoted wound closure, re-epithelialization and the expression of keratin 10 and keratin 14 in T2D mice on day 14. PLGA-rhFGF1 significantly decreased the levels of TNF- α and IL-6 in serum and skin tissues as well as the level of IL-1 β in skin tissues. Moreover, PLGA-rhFGF1 decreased the mRNA levels of *CXCL1*, *MCP1* and *MIP2*, and the fluorescence intensity of F4/80 and Ly6G in T2D wound mice and increased the collagen content and the protein expression of collagen I in the dermis of T2D wounds. PLGA-rhFGF1 also increased blood flow; the mRNA levels of the angiogenesis-related factors *VEGF*, *Ang-1* and *eNOS* and the fluorescence intensity of CD31 in T2D wound mice. These data indicate that PLGA releases rhFGF1 slowly and promotes skin wound healing in T2D mice. The mechanisms through which PLGA-rhFGF1 produces these effects involve decreased inflammation and the promotion of granulation, re-epithelialization and angiogenesis.

Keywords Diabetic wound healing, rhFGF1, PLGA, Nanofiber, Sustained-release system

Introduction

Diabetes, especially type 2 diabetes (T2D), is a metabolic disorder that is characterized by long-term high blood glucose levels and severely affects human health. Non-healing wounds are among the common complications of diabetes [1, 2]. Statistically, delayed healing of wounds occurs in approximately 25% of diabetes patients, and in severe cases, can lead to limb amputation and death [3]. The mortality rate within 5 years of amputation is as high as 70% [4]. At present, therapeutic methods for treating diabetic skin wounds include glycemic control, anti-infection measures, and surgery. However, these

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therapeutic methods mainly control the symptoms associated with diabetic wounds without curing the disease [5, 6]. Therefore, finding new drugs for the treatment of diabetic skin wounds has great scientific importance.

Growth factor-based therapies have emerged as promising approaches to accelerate wound healing. Among these factors, fibroblast growth factor 1 (FGF1), also known as acidic FGF (aFGF), has particular potential in diabetic wound healing [7–9]. Unlike vascular endothelial growth factor (VEGF), which primarily targets endothelial cells, or platelet-derived growth factor (PDGF), which mainly stimulates fibroblasts, FGF1 is a broad-spectrum mitogen that promotes the proliferation of fibroblasts, keratinocytes, and endothelial cells simultaneously [10, 11]. Furthermore, FGF1 can regulate glycolipid metabolism and suppress inflammation, which is theoretically ideal for the complex, multifaceted pathology of diabetic wounds [12–14]. However, the clinical translation of FGF1 has been severely hindered by its short half-life and instability; as a polypeptide, FGF1 is rapidly degraded by proteases in the wound environment [15]. For this reason, repeated administration of FGF1 is needed, which increases the cost of therapy. Therefore, reducing the cost of FGF1 therapy can facilitate the treatment of diabetic wounds.

To overcome these limitations, advanced drug delivery systems using nanotechnology have attracted significant attention [16, 17]. Nanofibers, particularly poly (lactic-co-glycolic acid) (PLGA) nanofibers, are resorbable biomaterials that have great potential for use as scaffolds in tissue engineering and as vehicles for the delivery of therapeutics [18]. The features of PLGA that have attracted attention include its high surface-area-to-volume ratio, biodegradability, biocompatibility, and feasibility for achieving sustained drug release and avoiding drug degradation [18, 19]. PLGA has promoted wound healing via encapsulated drugs such as antibiotics, anti-inflammatory drugs, proteins/peptides, and nucleic acids [20]. These reports suggest that PLGA can slowly deliver FGF1, thereby promoting diabetic wound healing. However, the effective application of a core-shell structure to encapsulate and protect the unstable proteins, such as FGF1, remains challenging. Traditional blending methods often expose proteins to organic solvents or result in an initial burst release [21]. Therefore, we prepared PLGA-loaded rhFGF1 (PLGA-rhFGF1) by electrospinning and tested its ability to treat diabetic skin injury. This research provides a theoretical basis for the application of fiber delivery systems in FGF-based drugs.

This study aimed to explore the therapeutic effects and mechanisms of PLGA-rhFGF1 nanofiber membranes in full-thickness skin wounds in diabetic mice. We found that PLGA promotes the sustainable release of FGF1 in a way that promotes diabetic wound healing and that the

therapeutic mechanism of PLGA-rhFGF1 is associated with the regulation of re-epithelialization, granulation tissue formation, inflammation and angiogenesis.

Materials and methods

Reagents and antibodies

RhFGF1 was obtained from *Escherichia coli* as described previously [22]. Primary antibodies against keratin 10 (Cat. ab76318), keratin 14 (Cat. ab181595) and Ly6G (Cat. ab25377) were purchased from Abcam (MA, USA). Collagen I (Cat. 14695-1-AP), and CD31 (Cat. AF3628) and F4/80 (Cat. sc-377009) were purchased from Proteintech Biotechnology (Hubei, China), R&D Systems (MN, USA) and Santa Cruz Biotechnology (TX, USA), respectively.

Preparation of the PLGA-rhFGF1 nanofiber membrane sustained-release system

PLGA and PLGA-loaded rhFGF1 nanofibers were prepared using coaxial electrospinning technology. Briefly, 10 mg of lyophilized rhFGF1 powder was dissolved in 1 mL of PBS to prepare a core layer solution with a concentration of 10 mg/mL. For the shell solution, PLGA was dissolved in hexafluoro-isopropyl alcohol (HFIP) under magnetic stirring for 2 h at room temperature to obtain a 15% (w/v) solution. The rhFGF1 solution and PLGA solution were loaded into a syringe. Next, the syringes were connected to the needle (model 18/25 G). PLGA-rhFGF1 nanofibers were obtained by starting the coaxial electrospinning power supply. The electrospinning process was conducted under the following parameters: a supply voltage of 20 kV, a tip-to-collector distance of 12 cm, flow rates of 0.0003 mm/s for the core and the shell, an ambient humidity of 70%, a temperature of 25 °C, and a spinning duration of 1 h. PLGA-rhFGF1 nanofibers were placed in a freeze dryer for 24 h before being stored at -80 °C.

Micromorphological observation of PLGA-rhFGF1 fibers

PLGA and PLGA-rhFGF1 were fixed on aluminum plates with conductive tape. After gold-plating of the aluminum plate, the micromorphology of the PLGA and PLGA-rhFGF1 nanofibers was observed by scanning electron microscopy (Hitachi, Tokyo, Japan). The diameters of the PLGA and PLGA-rhFGF1 nanofibers were measured by Nano Measurer software.

Detection of the core-shell structure

To verify the successful preparation of the PLGA-rhFGF1 nanofibers, the core-shell structure was examined. PLGA solution and FITC-labelled rhFGF1 (FITC-rhFGF1) solution were attached to the shell and core layers of the sprinkler head, respectively. Next, the PLGA-loaded with FITC-rhFGF1 nanofibers were obtained by

electrospinning technology. The shell and core layers of PLGA-rhFGF1 were observed via fluorescence microscopy using a bright field and a 488 nm laser.

Infrared spectroscopic determination of the PLGA-rhFGF1 composition

To investigate whether the structural characteristics of PLGA changed after rhFGF1 was loaded into the PLGA nanofibers, the compositions of rhFGF1, PLGA and PLGA-rhFGF1 were examined by infrared spectroscopy.

Hydrophilicity testing

The contact angle was used to evaluate the hydrophilicity of the fibers. The PLGA and PLGA-rhFGF1 and PLGA membranes were cut into 20 mm × 50 mm pieces and pasted onto flat slides. Changes in the contact angle of the sample that occurred within 1 min were measured using a contact angle meter.

Determination of the mechanical properties of the PLGA-rhFGF1 fibers

The PLGA-rhFGF1 and PLGA membranes were cut into pieces 0.5 cm × 2.0 cm pieces and stretched to 1 cm. The samples were placed on the mechanical instrument, and the stretching speed was set to 10 mm/min. The sample thickness (0.02 mm) was measured with a Vernier caliper and recorded.

In vitro release of rhFGF1 from PLGA fibers

The rhFGF1 nanofibers were transferred to tubes, and approximately 2 mL of PBS buffer solution was added. The tube-loaded rhFGF1 nanofibers were placed on a constant-temperature shaking table at 37 °C for 15 days. On days 1, 3, 5, 7, 9, 12 and 15, the supernatant containing rhFGF1 was collected and replaced with an equal volume of fresh PBS to maintain the total volume. The release of rhFGF1 was measured with an ELISA kit (Boster, Wuhan, China). The final mass of rhFGF1 at each time point was the sum of the mass at the specified time point and the mass at all the previous time points.

Cell culture

Human immortalized keratinocyte (HaCaT) cells were purchased from BIOBW (Beijing, China). The cells were cultured in modified Eagle's medium (MEM) supplemented with 15% fetal bovine serum (FBS) (Gibco, Grand Island, NE, USA) and 1% penicillin–streptomycin in a humidified atmosphere at 37 °C with 5% CO₂.

Evaluation of the in vitro biological safety by cell viability

We chose HaCaT cells to evaluate in vitro biological safety. The reason for choosing HaCaT cells is that epidermal cells are composed mainly of keratinocytes. Two different methods were adopted to evaluate safety.

First, the cytotoxicity of the released rhFGF1 and PLGA nanofibers on HaCaT cells was evaluated. The PLGA and PLGA-rhFGF1 membranes were sterilized by ultraviolet irradiation for 2 h. Next, the PLGA and PLGA-rhFGF1 membranes were incubated in centrifuge tubes containing 3 mL of MEM and placed in a constant-temperature shaker at 37 °C and 100 rpm/min for 48 h. Then, MEM was collected as the original extract solution for further study. HaCaT cells were seeded into 96-well plates at a density of 1×10^4 cells/well and cultured for 24 h. Then, the cells were treated with diluted original extracts of different concentrations (1, 1/2 and 1/4) for 24 h and 48 h. The cells were incubated with 200 µL of 0.5 mg/mL MTT solution at 37 °C for 3 h. The supernatant was discarded, and the formazan was dissolved in 150 µL of dimethyl sulfoxide (DMSO). The optical density (A value) was measured at a wavelength of 490 nm using a microplate reader (BioTek, Winooski, VT, USA). Cell viability was calculated using the following equation: cell viability (%) = $1 - (A_{\text{Ctrl}} - A_{\text{Sample}}) / (A_{\text{Ctrl}} - A_{\text{Blank}}) \times 100\%$.

We also investigated the effect of direct contact between PLGA and PLGA-rhFGF1 on cell viability. HaCaT cells were cultured on PLGA and PLGA-rhFGF1. The PLGA and PLGA-rhFGF1 membranes were placed in 48-well plates and sterilized by ultraviolet irradiation for 2 h. Next, HaCaT cells were cultured in these sterilized PLGA and PLGA-rhFGF1 membranes at a density of 2.5×10^4 cells/well for 24 h and 48 h. The viability of HaCaT cells was also detected with an MTT assay.

Establishment of a diabetic skin wound model

Male C57BL/6 mice were obtained from the Animal Center of the Chinese Academy of Sciences (Beijing, China). All animal procedures were approved by the Animal Research Ethics Committee of Wenzhou Medical University. A type 2 diabetes (T2D) model was induced by feeding the mice a high-fat diet (60% of the calories were derived from fat, Research Diets, New Brunswick, NJ, United States) for 3 months, followed by intraperitoneal injections of streptozotocin (STZ, 35 mg/kg) for 5 consecutive days.

Mice with blood glucose concentrations > 16.7 mmol/L were considered diabetic and used for experiments. Twenty-eight diabetic mice were randomly divided into four groups: control (untreated, Ctrl), PLGA alone, rhFGF1 alone, and PLGA-rhFGF1. Under isoflurane anesthesia, the hair on the dorsum area was shaved with an electric clipper, and depilatory creams were used to clear up the residual hair. Two silicone-splinted rings with a 0.5-mm thickness, 16-mm external diameter, and 8-mm internal diameter were fixed on either side of the wounds using 6-0 nylon sutures (Lingqiao, Zhejiang, China). A 6-mm full-thickness excisional wound was created on the dorsal skin and splinted with a silicone

ring. Each mouse had 2 wounds on the back and 1 on each side ($n=7$ mice; 14 wounds/group). The wounds in each group were treated only once with saline (Ctrl group), PLGA (PLGA group), rhFGF1 (rhFGF1 group), and PLGA-rhFGF1 (PLGA-rhFGF1 group). The dose of rhFGF1 in both the PLGA-rhFGF1 group and the rhFGF1 group was 2 μg . Next, all the experimental mice were treated with a Tegaderm transparent dressing (3 M Health Care, Maplewood, MN, USA) to avoid infection and were wrapped with self-adhesive bandages (Maxsoins, Zhejiang, China) to prevent chewing of the splints. Wounds were monitored photographically on days 0, 3, 7, and 14 and measured using Image-Pro Plus (Media Cybernetics, Rockville, MD, USA). The wound closure rate was calculated as follows: $(\text{initial area} - \text{current area}) / \text{initial area} \times 100\%$. The mice were sacrificed on days 7 and 14, and blood and skin tissue samples were collected for further examination. Serum was obtained from blood by centrifugation at 3500 rpm for 10 min at 4 °C and then stored at -20 °C. Skin tissues were stored at -80 °C for RT-PCR or fixed with 4% paraformaldehyde for histopathological, immunohistochemical and immunofluorescence staining.

Paraffin-embedded sections of skin tissue

The skin tissues were fixed overnight with 4% paraformaldehyde, dehydrated with gradient alcohol (85% ethanol for 1 h, 95% ethanol twice for 1 h, and 100% ethanol twice for 30 min), incubated with xylene twice for 15 min, embedded in paraffin for 2 h and 1.5 h and sectioned into 5- μm -thick sections.

Histopathological staining

Hematoxylin and eosin (HE) (Solarbio, Beijing, China) and Masson's trichrome (Solarbio, Beijing, China) were used to evaluate diabetic wound closure and collagen formation. Before HE staining and Masson's trichrome staining, the skin tissue sections were dewaxed with xylene twice for 15 min and hydrated with gradient alcohol (100% ethanol twice for 5 min, 95% ethanol for 5 min, 85% ethanol for 5 min, 75% ethanol for 5 min and purified water three times for 5 min each). Next, the skin tissue sections were subjected to HE staining and Masson's trichrome staining.

The HE staining procedure was as follows: the skin tissue sections were stained with hematoxylin solution for 5 min, washed with tap water for 8 min, stained with eosin solution for 1 min, washed with pure water three times for 5 min each, dehydrated with gradient alcohol (85% ethanol for 2 min, 95% ethanol 2 min, and 100% ethanol twice for 2 min), incubated with xylene twice for 5 min each, sealed with neutral resin, captured using a light microscope (Nikon, Tokyo, Japan) and analyzed

with ImageJ (NIH) software (National Institutes of Health, Bethesda, MD, USA).

Masson's trichrome staining procedure was as follows: the skin tissue sections were stained with iron hematoxylin solution for 10 min, washed with purified water for 5 min, differentiated with acidic ethanol solution for 5 s, washed with purified water for 5 min, stained with Masson solution for 5 min, washed with purified water for 5 min, stained with Lichun red and Fuchsin solution for 1 min, washed with 0.2% acetic acid solution for 5 min, washed with phosphomolybdic acid solution for 2 min, washed with 0.2% acetic acid solution for 5 min, stained with aniline blue solution for 2 min, washed with 0.2% acetic acid solution for 5 min, dehydrated with gradient alcohol (85% ethanol for 2 min, 95% ethanol 2 min, and 100% ethanol twice for 2 min), incubated with xylene twice for 5 min each time, sealed with neutral resin, captured using a light microscope (Nikon, Tokyo, Japan) and analyzed with ImageJ (NIH) software (National Institutes of Health, Bethesda, MD, USA).

Immunohistochemical and immunofluorescence staining

For immunohistochemical staining, skin tissue sections were deparaffinized, rehydrated, subjected to antigen retrieval in citrate buffer for 5 min, blocked with 5% bovine serum albumin (BSA) for 1 h at room temperature, and incubated overnight at 4 °C with primary antibodies against keratin 10 (1:500 dilution, Abcam), keratin 14 (1:500, Abcam), or collagen I (1:500, Proteintech). After being washed, the sections were incubated with HRP-conjugated secondary antibodies (1:500, Abcam) for 1 h at room temperature, visualized with DAB substrate for 5 min, stained with hematoxylin solution for 5 min, washed with tap water for 8 min, stained with eosin solution, dehydrated with gradient alcohol, incubated with xylene, sealed with neutral resin and captured using a light microscope (Nikon, Tokyo, Japan).

For immunofluorescence staining, skin tissue sections were deparaffinized, rehydrated, subjected to antigen retrieval in citrate buffer for 5 min, blocked with 5% bovine serum albumin (BSA) for 1 h at room temperature, and incubated overnight at 4 °C with primary antibodies against CD31 (1:300, R&D Systems), F4/80 (1:300, Santa Cruz), or Ly6G (1:300, Abcam) overnight. Following the PBS washes, the sections were incubated with fluorophore-conjugated secondary antibodies (1:500) for 1 h in the dark. The nuclei were counterstained with DAPI (1:1000). Images were acquired using a laser scanning confocal microscope (Nikon, Japan), and the fluorescence intensity was quantified using ImageJ software (NIH, USA).

Measurement of pro-inflammatory cytokine levels by enzyme-linked immunosorbent assay (ELISA) assay

The sample was equilibrated at room temperature and diluted to the appropriate concentration (TNF- α and IL-6 were diluted 3 times, while IL-1 β was not diluted). The concentrations of IL-1 β , TNF- α and IL-6 in mouse serum were measured by ELISA following the manufacturer's directions (EIAab, Hubei, China). The standard and sample (100 μ L) to be tested were added to the well plate. The plate was incubated on an oscillator at 37 $^{\circ}$ C for 2 h. After the liquid in the well plate was discarded, the primary antibody (100 μ L) was incubated at 37 $^{\circ}$ C for 1 h. The plate was subsequently washed 3 times with washing buffer. The secondary antibody (100 μ L) was incubated at 37 $^{\circ}$ C for 1 h, and the plate was subsequently washed 3 times with washing buffer. The substrate (90 μ L) was added to the wells, and the color was developed at 37 $^{\circ}$ C for 20 min. The reaction was terminated by the addition of a stop solution. The absorbance was read at a wavelength of 450 nm using a microplate reader, and the levels of pro-inflammatory cytokines were calculated using the curve formula.

Reverse transcription polymerase chain reaction (RT-PCR)

TRIzol reagent (Invitrogen, Carlsbad, CA, USA) was used to extract total RNA from skin tissue samples. To synthesize cDNA, 0.5 μ g of total RNA was reverse transcribed at 42 $^{\circ}$ C for 2 min, 37 $^{\circ}$ C for 15 min and 85 $^{\circ}$ C for 5 s using a PrimeScript[™] RT reagent kit with gDNA Eraser (Perfect Real Time) (TaKaRa, Japan). PCR was performed using a SYBR green-based RT-PCR kit (TaKaRa, Japan)

Table 1 Primers sequences for RT-PCR

Gene	Primer sequences
<i>TNFα</i>	5'-CAGGCGGTGCCTATGTCTC-3' 5'-CGATCACCCGAAGTTCAGTAG-3'
<i>IL-1β</i>	5'-AAATACCTGTGCCTTGGGC-3' 5'-CTTGGGATCCACACTCTCCAG-3'
<i>IL-6</i>	5'-AGAAGGAGTGGCTAAGGACCAA-3' 5'-AACGCACTAGGTTTGCCGAGTA-3'
<i>CXCL1</i>	5'-CTGGGATTACCTCAAGAATC-3' 5'-CAGGGTCAAGGCAAGCCTC-3'
<i>MIP2</i>	5'-CCAACCACCAAGGCTACAGG-3' 5'-GCGTCACACTCAAGCTCTG-3'
<i>MCP1</i>	5'-TTAAAAACCTGGATCGGAACCAA-3' 5'-GCATTAGCTTCAGATTTACGGGT-3'
<i>Ang-1</i>	5'-CACATAGGGTGCAGCAACCA-3' 5'-CGTCGTGTTCTGGAAGAATGA-3'
<i>Ang-2</i>	5'-AGAATAAGCAAGTCTCGCTTCC-3' 5'-TGAACCCCTTAGAGGCTCGGT-3'
<i>eNOS</i>	5'-TCAGCCATCACAGTGTTCCTCC-3' 5'-ATAGCCCGCATAGCGTATCAG-3'
<i>VEGF</i>	5'-GCACATAGAGAGAATGAGCTTCC-3' 5'-CTCCGCTCTGAACAAGGCT-3'
<i>ACTIN</i>	5'-CCTCACTGTCCACCTTCC-3' 5'-GGGTGTAAACGCAGCTC-3'

on a CFX Connect Real Time PCR detection system (Bio-Rad, Hercules, CA, USA). After an initial denaturation step at 95 $^{\circ}$ C for 2 min, 40 cycles of PCR were carried out. Each cycle consisted of a melting step at 95 $^{\circ}$ C for 5 s and an annealing extension step at 60 $^{\circ}$ C for 30 s. The data were quantified by the 2^{- $\Delta\Delta$ Ct} method and normalized to ACTB expression. The primers (GenePharma, Suzhou, China) used for RT-PCR are detailed in Table 1.

Doppler blood flow measurement

Angiogenesis in diabetic wounds was evaluated by measuring blood flow. Blood perfusion in the injured regions of the mice was measured at 14 days using a laser Doppler fiber (Perimed, Stockholm, Switzerland).

Statistical analysis

Statistical data were analyzed via one-way analysis of variance (ANOVA) followed by Tukey's post-hoc test using GraphPad Prism version 9.0 software. All the data are presented as the means $\pm$ SD. $P < 0.05$ was considered to indicate statistical significance.

Results

Characterization and release of PLGA and PLGA-rhFGF1

SEM revealed that the PLGA and PLGA-rhFGF1 fibers were continuously formed and that their surfaces were smooth. The diameter of the PLGA fibers was 844.33 $\pm$ 161.01 nm, and the diameter of the PLGA-rhFGF1 fibers was 732.33 $\pm$ 151.51 nm (Fig. 1A). The diameter of the PLGA-rhFGF1 fibers was smaller than that of the PLGA fibers, mainly because the addition of rhFGF1 to the PLGA fibers reduced the viscosity of the fibers, making them easier to stretch and thus thinner. To verify the successful preparation of the PLGA-rhFGF1 nanofibers, the core shell structure was examined. We found that FITC-rhFGF1 was present in the core layer and that the rhFGF1 was surrounded by a shell composed of PLGA (Fig. 1B). The characteristic absorption peaks of PLGA-rhFGF1 included the characteristic absorption peaks of PLGA (1745 cm⁻¹ and 1053 cm⁻¹) and rhFGF1 (3422 cm⁻¹ and 1627 cm⁻¹), indicating that PLGA successfully encapsulated rhFGF1 (Fig. 1C). The hydrophilic properties of PLGA and PLGA-rhFGF1 were evaluated by measuring changes in the contact angle. The contact angles of PLGA and PLGA-rhFGF1 were 137.71 $\pm$ 1.92 $^{\circ}$ and 141.29 $\pm$ 1.27 $^{\circ}$, respectively. The data indicated no significant difference in the hydrophilicity of the PLGA and PLGA-rhFGF1 nanofibers (Fig. 1D). Moreover, the tensile strength of the PLGA-rhFGF1 fibers was 5.52 $\pm$ 0.55 MPa, and their fracture productivity was 171.89 $\pm$ 17.70%. The tensile strength of PLGA was 9.33 $\pm$ 1.00 MPa, and its fracture productivity was 335.58 $\pm$ 82.91% (Fig. 1E). The fiber diameter can influence the stress, and because PLGA-rhFGF1 fibers are thinner than PLGA fibers, the

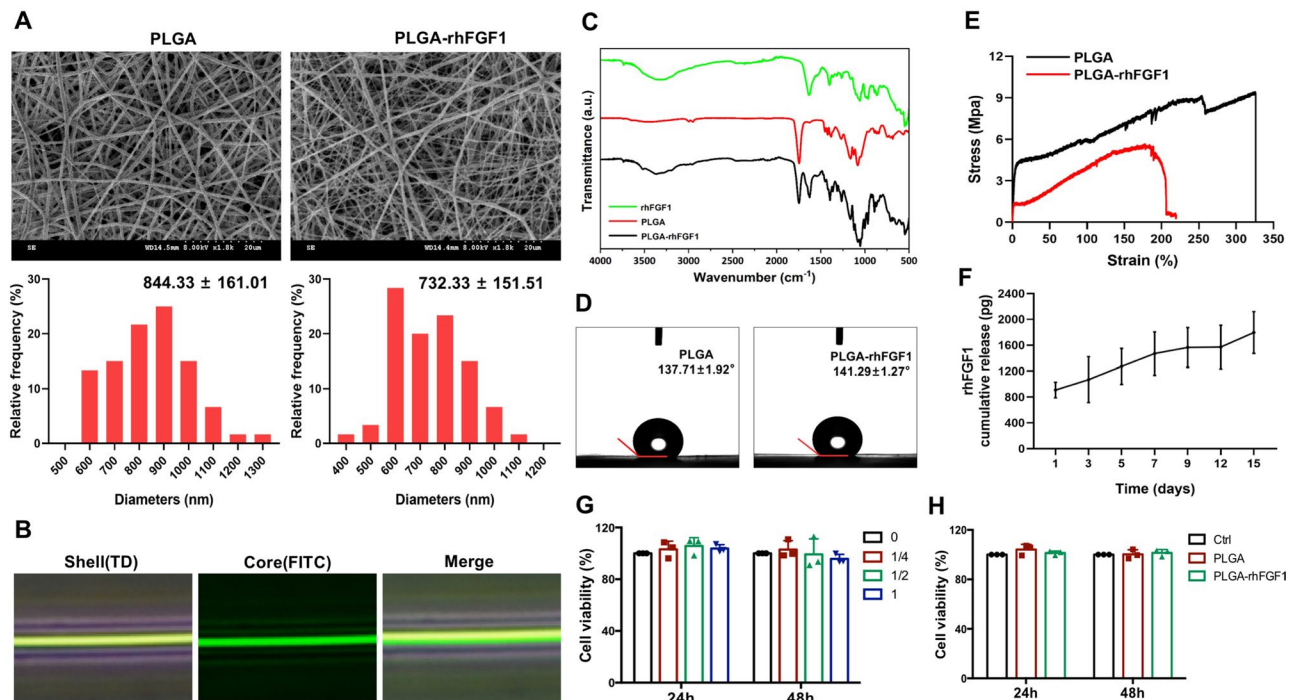


Fig. 1 Characterization and release of PLGA and PLGA-rhFGF1. **(A)** The micromorphological structures of PLGA and PLGA-rhFGF1 were observed by scanning electron microscopy, and the diameters of the PLGA and PLGA-rhFGF1 nanofibers were measured using Nano Measurer software. **(B)** The core-shell structure of PLGA-rhFGF1 was observed via bright field microscopy and under a 488-nm laser via fluorescence microscopy. RhFGF1 was labelled with FITC. **(C)** Infrared spectroscopic evaluation of PLGA-rhFGF1. **(D)** The hydrophilic properties of PLGA and PLGA-rhFGF1 were evaluated by measuring changes in the contact angle. **(E)** Mechanical properties of PLGA and PLGA-rhFGF1. **(F)** In vitro release of rhFGF1 from PLGA nanofibers at 37 °C over a period of 15 days. **(G, H)** The cell viability of HaCaT cells was detected by MTT assay. The data are presented as the means ± SD

tensile strength of PLGA-rhFGF1 is lower than that of PLGA. Although the mechanical properties of the PLGA and PLGA-rhFGF1 fibers differ, fibers of both compositions have good mechanical properties and possess the potential to be used as biological dressings.

Next, measurement of the in vitro release of rhFGF1 from the PLGA-rhFGF1 nanofibers revealed that rhFGF1 was released during the 15-day experimental period. RhFGF1 was released most rapidly on the first day, followed by slow release, as shown by the coaxial electrospinning release curve (Fig. 1F). In addition, we found that the released rhFGF1 by the PLGA-rhFGF1 membranes did not affect cell survival and that the PLGA and PLGA-rhFGF1 membranes did not affect the viability of HaCaT cells when in direct contact. The results suggested that PLGA and rhFGF1 are safe. Therefore, these results show that PLGA is a good sustained-release material for enhancing the stability and bioactivity of rhFGF1.

PLGA-rhFGF1 promoted wound healing in T2D mice

A flow diagram of the experiments conducted in this study is shown in Fig. 2A. Compared with Ctrl group (6.8%±8.1; 23.1%±9.1; 74.5%±3.7), T2D mice in the PLGA (13.5%±6.7; 35.3%±8.5; 67.0%±9.7) and rhFGF1 (12.5%±9.1; 38.9%±9.8; 77.0%±4.0) groups did not exhibit

increased wound closure on day 3, 7 or 14. However, the T2D mice in the PLGA-rhFGF1 group (90.1%±3.2) displayed significantly increased skin wound closure on day 14 (Fig. 2B and C). The results of HE staining further verified that PLGA-rhFGF1 (1192.3±297.2) and rhFGF1 (1815.1±370.3) effectively promoted diabetic wound closure on day 14 and that the effect of PLGA-rhFGF1 on wound closure was greater than that of rhFGF1. Compared with the Ctrl group (2561.0±344.2), treatment with PLGA (2621.1±329.9) alone did not have a therapeutic effect (Fig. 2D and E). These results show that PLGA-rhFGF1 promotes wound healing in T2D mice.

PLGA-rhFGF1 promoted re-epithelialization in T2D wound mice

The effects of PLGA-rhFGF1 on re-epithelialization were evaluated by HE staining. Compared with those in the Ctrl group (75.4 ± 14.7), rhFGF1 (97.1 ± 6.5) and PLGA-rhFGF1 (98.2 ± 4.0) promoted re-epithelialization in T2D wound mice on day 14 (Fig. 3A). Re-epithelialization is associated with epidermal differentiation and proliferation, of which keratin 10 and keratin 14, respectively, are markers [23, 24]. Therefore, the expressions of keratin 10 and keratin 14 in the wound healing location were measured by immunohistochemical staining. PLGA (8.8 ±

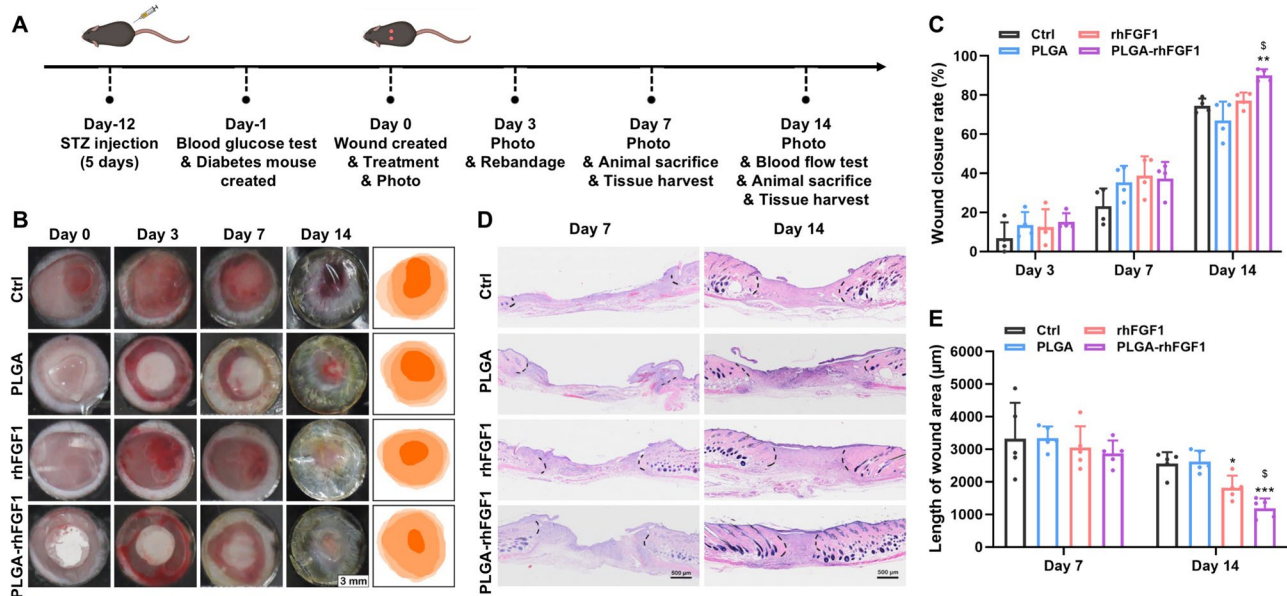


Fig. 2 PLGA-rhFGF1 promotes wound healing in type 2 diabetic (T2D) mice. **(A)** Flow diagram of the experiment. **(B, C)** Representative images showing PLGA-rhFGF1-promoted diabetic wound healing and quantification of the wound closure rate in diabetic wound mice ($n=5$). **(D, E)** HE staining of the wound areas in diabetic wound-bearing mice on days 7 and 14. Quantification of diabetic wound healing ($n=5$). The data are presented as the means $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ versus diabetic wound model mice (Ctrl); $^{\S}P < 0.05$ versus rhFGF1-treated diabetic skin wound model mice (rhFGF1)

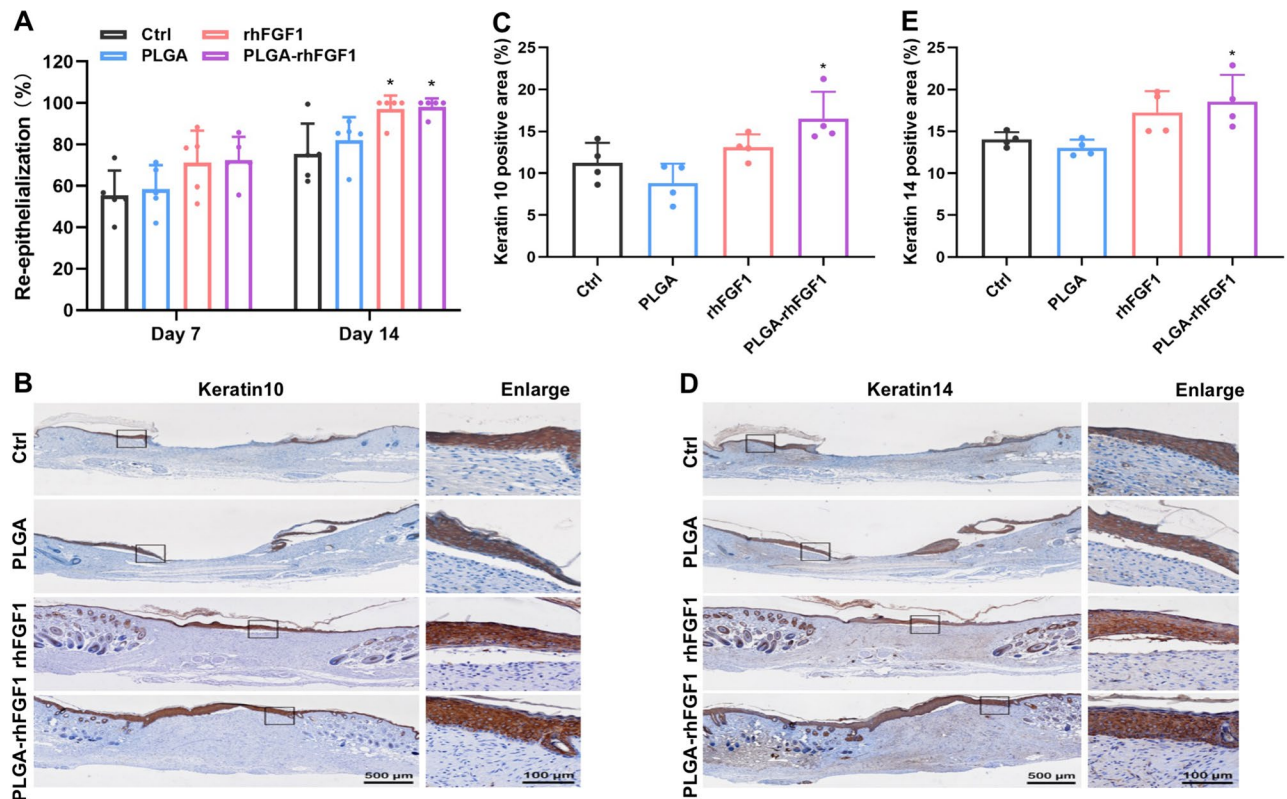


Fig. 3 PLGA-rhFGF1 promoted re-epithelialization in T2D wound mice. **(A)** Quantification of re-epithelialization according to the results of HE staining on days 7 and 14 ($n=5$). **(B, C)** The protein expression of keratin 10 was detected by immunohistochemical staining. Quantification of keratin 10 protein levels ($n=4$). **(D, E)** The protein expression of keratin 14 was detected by immunohistochemical staining. Quantification of keratin 14 protein levels ($n=4$). The data are presented as the means $\pm$ SD. * $P < 0.05$ versus diabetic wound-bearing mice (Ctrl)

2.3; 13.0 ± 1.0) or rhFGF1 (13.1 ± 1.5 ; 17.3 ± 2.5) treatment did not increase the expressions of keratin 10 or keratin 14 in T2D mice on day 14, compared with that in the Ctrl group (11.2 ± 2.4 ; 14.0 ± 0.9). However, treatment with PLGA-rhFGF1 (16.5 ± 3.2 ; 18.5 ± 3.2) increased the expression of both proteins in T2D mice on day 14 (Fig. 3B-E). These results suggest that PLGA-rhFGF1 promotes re-epithelialization in T2D mice by regulating epidermal proliferation and differentiation.

PLGA-rhFGF1 suppressed inflammation in T2D wound mice

Delayed diabetic wound healing aggravates the inflammatory response through the release of pro-inflammatory cytokines [25]. Therefore, the levels of IL-1 β , TNF- α and IL-6 in serum and skin tissues were detected by ELISA and RT-PCR, respectively. PLGA treatment did not decrease the levels of IL-1 β (106.4 ± 8.9 , 1.05 ± 0.4), TNF- α (35.1 ± 2.9 , 0.74 ± 0.2) and IL-6 (68.2 ± 13.2 , 0.79 ± 0.4) in the serum or skin tissue, and rhFGF1 treatment did not decrease the serum levels of IL-1 β (92.7 ± 14.9) or IL-6 (52.8 ± 8.6) in skin tissue. However, PLGA-rhFGF1 significantly decreased the levels of TNF- α (29.8 ± 2.6 , 0.29 ± 0.08) and IL-6 (53.4 ± 5.5 , 0.51 ± 0.2) in serum and skin tissue as well as the levels of IL-1 β (0.30 ± 0.08) in skin tissues (Fig. 4A-F). Moreover, compared with that of rhFGF1, the inhibitory effect of PLGA-rhFGF1 on TNF- α and IL-6 in skin tissues was stronger (Fig. 4D and E).

These results indicated that PLGA-rhFGF1 suppressed inflammatory responses in T2D wound mice.

PLGA-rhFGF1 inhibited immune cell-mediated inflammation in T2D wound mice

Macrophages and neutrophils, the primary immune cells in peripheral blood, migrate to locations at which diabetic damage occurs [26, 27]. Therefore, the infiltration of skin wounds in T2D mice by macrophages and neutrophils was evaluated by RT-PCR and immunofluorescence. The presence of *MIP2*, *MCP1* and *F4/80* indicates macrophage infiltration, whereas the presence of *CXCL1* and *Ly6G* indicates neutrophil infiltration. We found that PLGA-rhFGF1 and rhFGF1 decreased the mRNA levels of *CXCL1* (0.30 ± 0.2 ; 0.66 ± 0.1), *MCP1* (0.33 ± 0.1 ; 0.54 ± 0.2) and *MIP2* (0.43 ± 0.1 ; 0.44 ± 0.2), but PLGA (1.16 ± 0.1 ; 0.70 ± 0.1 ; 0.82 ± 0.2) did not influence the expression of these chemokines (Fig. 5A-C). Moreover, compared with those in the Ctrl group (63.6 ± 7.7 ; 63.1 ± 3.9), on day 7, the fluorescence intensities of Ly6G and F4/80 were lower in T2D wound mice that had been treated with PLGA-rhFGF1 (36.2 ± 5.8 ; 36.6 ± 4.2) and rhFGF1 (48.2 ± 4.4 ; 49.0 ± 5.6), and the inhibitory effect of PLGA-rhFGF1 on Ly6G levels in skin was greater than that of rhFGF1 (Fig. 5 D-G). These results show that PLGA-rhFGF1 suppresses macrophage- and neutrophil-mediated inflammation in T2D mice.

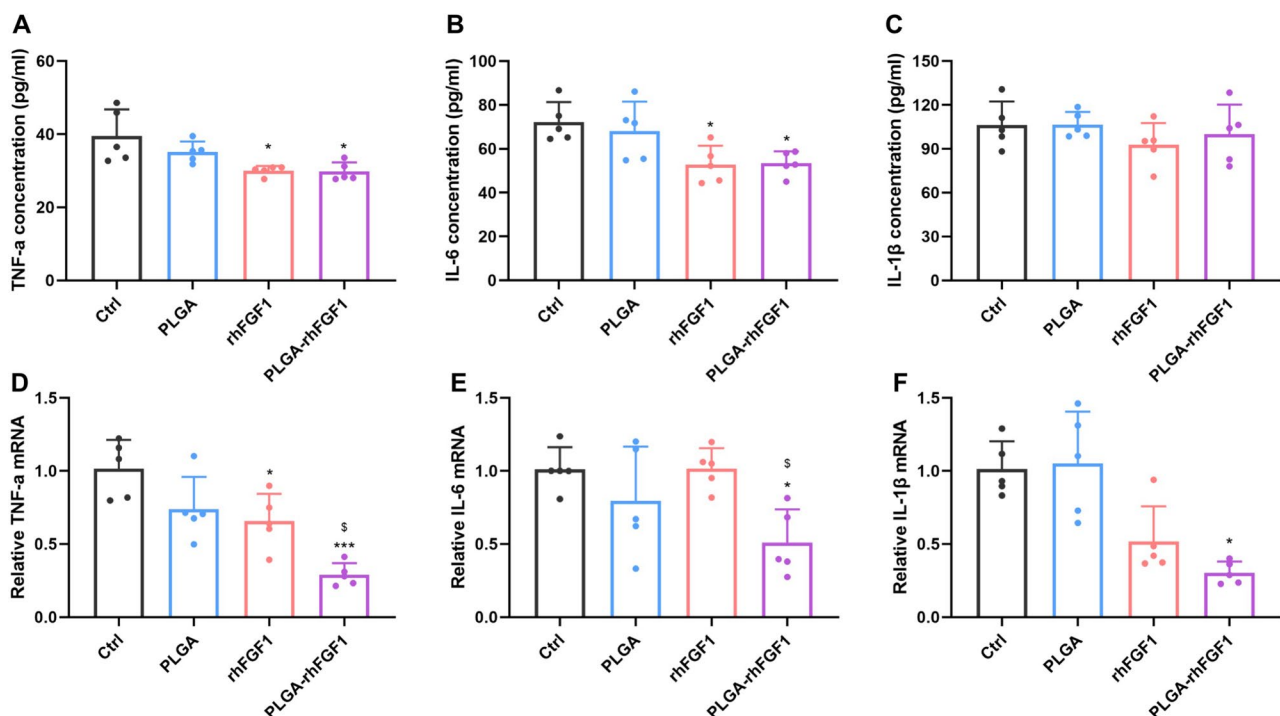


Fig. 4 PLGA-rhFGF1 fibers suppressed inflammation in T2D wound mice. (A-C) The concentrations of TNF- α , IL-6 and IL-1 β in diabetic mouse serum were determined by ELISA ($n=5$). (D-F) The mRNA levels of TNF- α , IL-6 and IL-1 β in diabetic mouse skin were measured via RT-PCR ($n=5$). The data are presented as the means $\pm$ SD. * $P < 0.05$, *** $P < 0.001$ versus diabetic wound model mice (Ctrl); $^{\S}P < 0.05$, $^{\S\S}P < 0.05$ versus rhFGF1-treated diabetic skin wound model mice (rhFGF1)

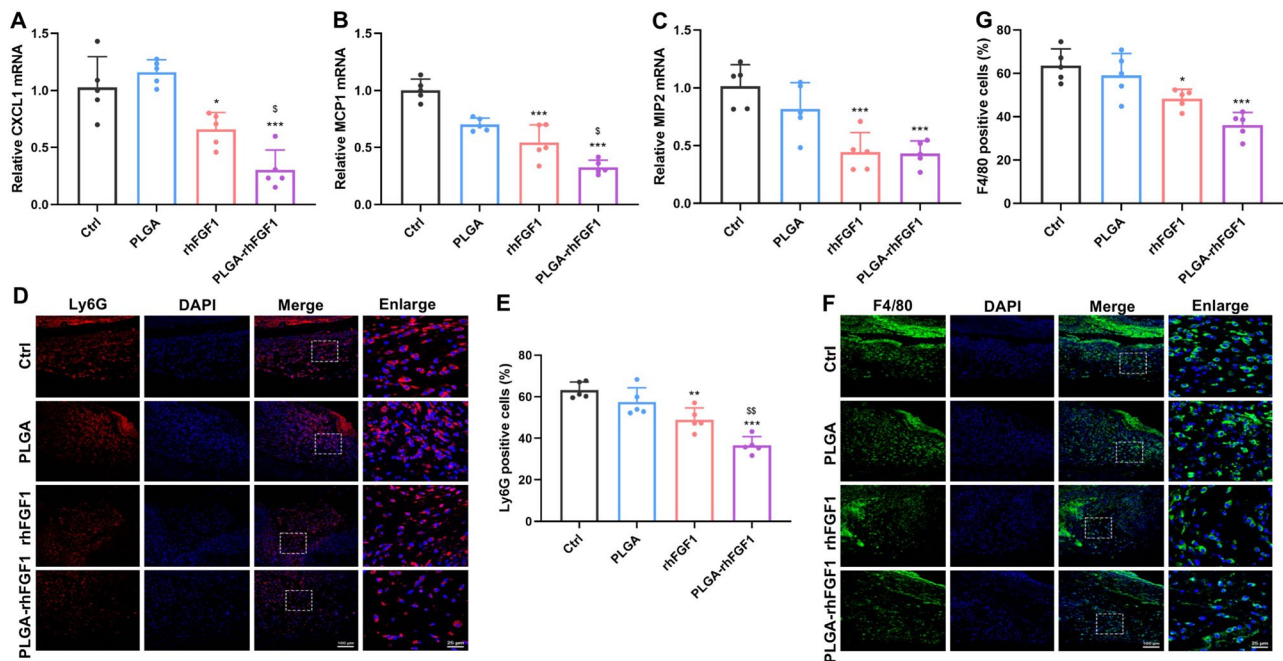


Fig. 5 PLGA-rhFGF1 inhibited immune cell-mediated inflammation in T2D wound mice. **(A–C)** The mRNA levels of *CXCL1*, *MCP1* and *MIP2* in diabetic mouse skin were measured via RT-PCR ($n=5$). **(D, E)** The fluorescence associated with Ly6G was detected and quantified by immunofluorescence staining and ImageJ software, respectively ($n=5$). **(F, G)** The fluorescence associated with F4/80 was detected and quantified by immunofluorescence staining and ImageJ software, respectively ($n=5$). The data are presented as the means $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ versus diabetic wound model mice (Ctrl); $^{\S}P < 0.05$, $^{\S\S}P < 0.01$ versus rhFGF1-treated diabetic skin wound model mice (rhFGF1)

PLGA-rhFGF1 promoted granulation in T2D wound mice

The formation of granulation tissue is associated with the collagen content of the dermis [28]. Therefore, the collagen content of diabetic wounds was measured by Masson's trichrome staining and immunohistochemistry. Compared with Ctrl group (9.2 ± 1.6 ; 28.7 ± 3.0), the PLGA treatment (7.7 ± 1.4 ; 27.7 ± 1.8) did not increase the collagen content of the skin of T2D wound mice on days 7 and 14, while the rhFGF1 treatment (34.8 ± 0.6) increased the collagen content of the skin of the T2D wound mice on day 14. PLGA-rhFGF1 treatment (13.0 ± 1.1 ; 39.1 ± 1.9) increased the collagen content of the skin of T2D wound mice on days 7 and 14, and the promoting effect of PLGA-rhFGF1 was greater than that of rhFGF1 at days 7 and 14 (Fig. 6A–C). In addition, PLGA-rhFGF1 (18.9 ± 1.6) and rhFGF1 (18.3 ± 1.6) treatment upregulated the protein expression of collagen I in T2D wound mice on day 14, but PLGA treatment did not increase the protein expression of collagen I (14.9 ± 1.6) (Fig. 6D and E). Therefore, these results revealed that PLGA-rhFGF1 granulation occurred in T2D wound mice.

PLGA-rhFGF1 promoted angiogenesis in T2D wound mice

Angiogenesis can accelerate diabetic wound healing [29]. Therefore, we first detected blood flow by a laser Doppler flowmeter. Compared with the Ctrl (2.13 ± 0.2) or rhFGF1 treatment (2.47 ± 0.3), the PLGA-rhFGF1 treatment (3.27 ± 0.7) increased blood flow in T2D wound

mice (Fig. 7A and B). Moreover, PLGA-rhFGF1 increased the mRNA levels of angiogenesis-related factors (*Ang-1*, 1.72 ± 0.7 ; *eNOS*, 2.3 ± 0.9 ; and *VEGF*, 2.26 ± 0.7) in diabetic wounds but did not affect *Ang-2* mRNA expression (0.98 ± 0.1) (Fig. 7C–F). CD31 is a marker of blood vessels; thus, we measured the protein expression of CD31 by immunofluorescence. Compared with the Ctrl (2.45 ± 0.3) and rhFGF1 (2.84 ± 0.2) groups, the PLGA-rhFGF1 (3.71 ± 0.4) group had increased CD31 fluorescence intensity in T2D wound mice (Fig. 7G, H). These results show that PLGA-rhFGF1 treatment promotes angiogenesis in T2D wound mice.

Discussion

Diabetes affects human health worldwide and presents challenges for clinical treatment. Effective management of diabetic wounds remains a clinical challenge because of the complex interplay among persistent inflammation, impaired angiogenesis, and defective matrix deposition [30]. In this study, we developed a core-shell PLGA nanofiber system for the sustained delivery of rhFGF1 and demonstrated its superior therapeutic efficacy in a T2D mouse wound model. Our findings indicate that by addressing key pathological barriers, PLGA-rhFGF1 significantly accelerates wound closure; PLGA-rhFGF1 suppresses prolonged inflammation and simultaneously promotes re-epithelialization and neovascularization.

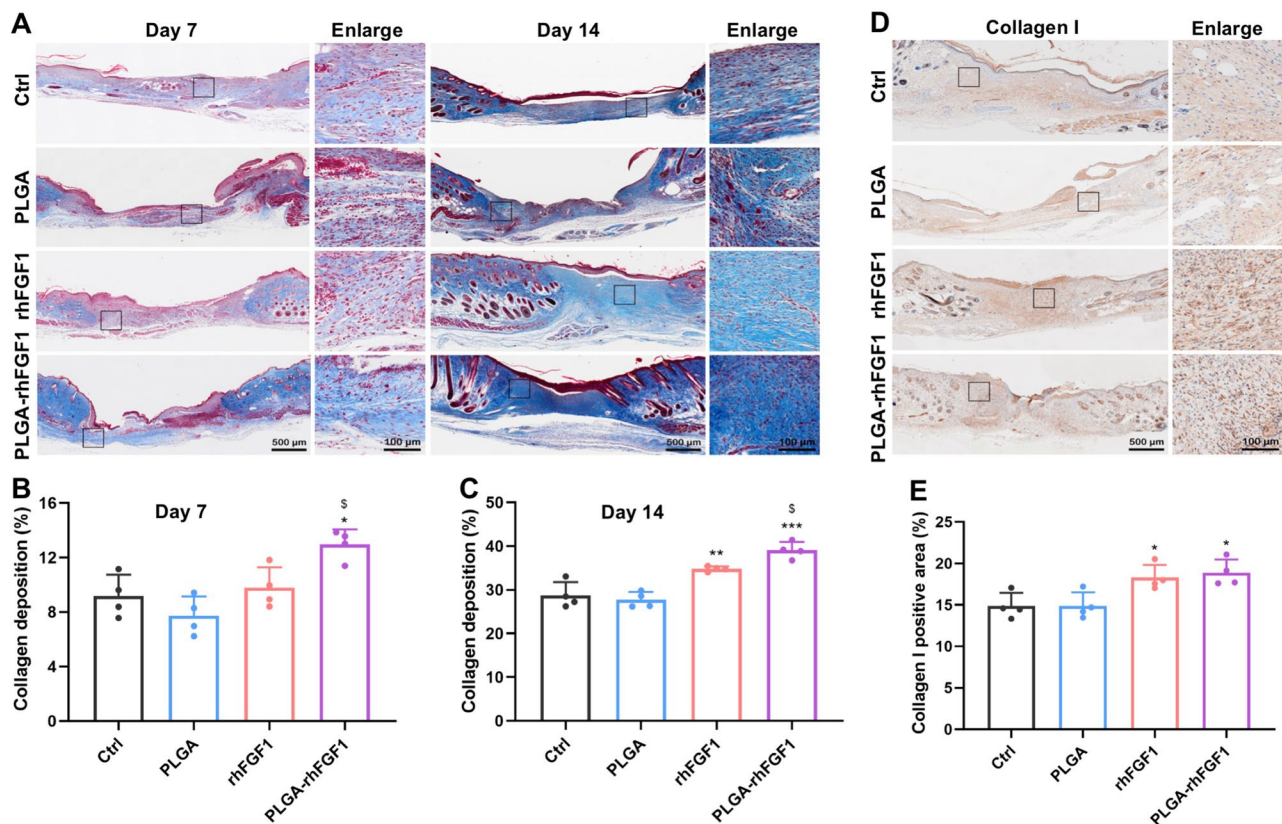


Fig. 6 PLGA-rhFGF1 promoted the formation of granulation tissue in T2D wound mice. **(A–C)** The effect of PLGA-rhFGF1 on granulation formation was evaluated by Masson staining. Quantification of the collagen content in T2D wound-bearing mice on days 7 and 14 ($n=4$). **(D, E)** The protein expression of collagen I in diabetic skin wounds was detected by immunofluorescence staining. Quantification of collagen I protein levels ($n=4$). The data are presented as the means $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ versus diabetic wound model mice (Ctrl); $^{\S}P < 0.05$ versus rhFGF1-treated diabetic skin wound model mice (rhFGF1)

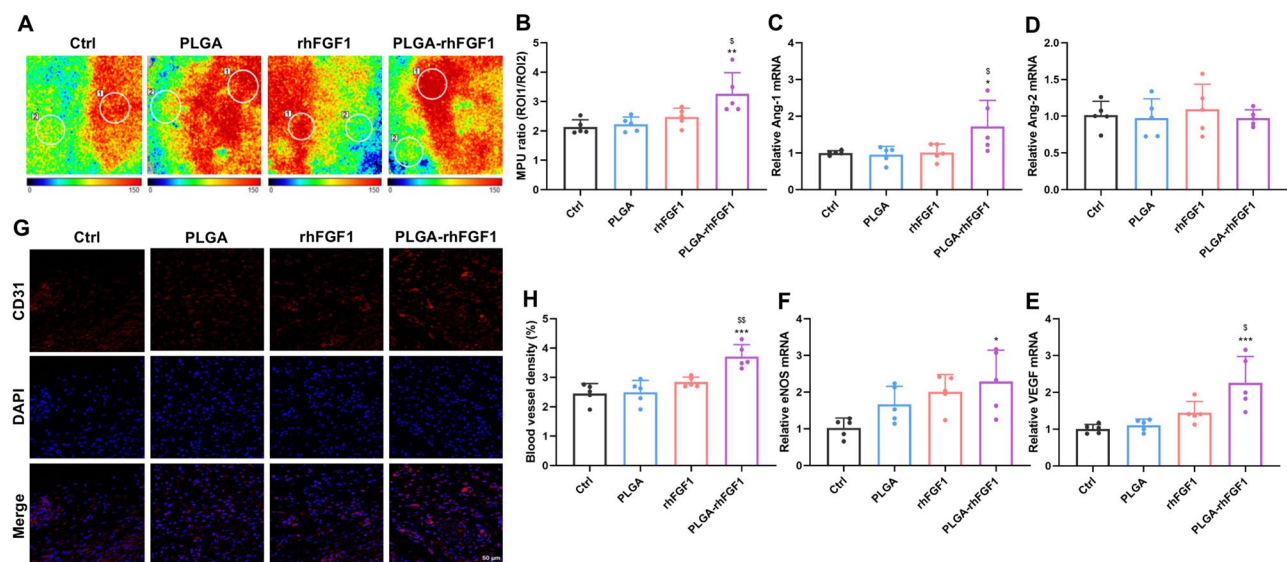


Fig. 7 PLGA-rhFGF1 promoted angiogenesis in T2D wound mice. **(A, B)** The effects of PLGA-rhFGF1 on blood flow were measured using a laser Doppler flowmeter. Quantification of the blood flow in T2D wound-bearing mice on day 14 ($n=5$). **(C–F)** The mRNA levels of *Ang-1*, *Ang-2*, *eNOS* and *VEGF* in diabetic mouse skin were measured via RT-PCR ($n=5$). **(G, H)** The protein expression of CD31 was measured by immunofluorescence staining and quantified using ImageJ software ($n=5$). The data are presented as the means $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ versus diabetic wound model mice (Ctrl); $^{\S}P < 0.05$, $^{\S\S}P < 0.01$ versus rhFGF1-treated diabetic skin wound model mice (rhFGF1)

FGF1 is a tissue repair factor that has various biological activities and functions [10]. However, FGF1 is a polypeptide and thus has the disadvantages of a short half-life, instability at room temperature and low utilization [15]. To overcome the shortcomings of FGF1 as a therapeutic agent, an increasing number of materials that can be used to achieve the sustained release of drugs have been developed in recent years [31, 32]. PLGA is a useful synthetic nanofiber that has high biocompatibility and good mechanical properties [33, 34]. Although PLGA is a potential sustained-release material for treating diabetic wounds, the effect of PLGA loaded with rhFGF1 on diabetic wound injury is unknown. A critical finding of our study is the enhanced stability and bioactivity of rhFGF1 when it is encapsulated in the core-shell structure. Unlike free FGF1, which is susceptible to rapid proteolytic degradation, our in vitro release data confirmed the sustained release of rhFGF1 for more than 14 days. This sustained availability is likely the primary driver for the superior wound healing outcomes observed in the PLGA-rhFGF1 group compared with those observed in the free rhFGF1 group on day 14. These findings are consistent with those of previous reports suggesting that the continuous delivery of growth factors is essential to match the physiological phases of tissue repair [35, 36].

Wound healing is related to decreased inflammation and the promotion of granulation, re-epithelialization and angiogenesis [25, 37, 38]. Inflammation is a double-edged sword in wound healing; although necessary for the initial phase of healing, chronic inflammation is characteristic of nonhealing diabetic wounds [25]. We observed that PLGA-rhFGF1 treatment significantly reduced the expression of pro-inflammatory cytokines (*TNF- α* , *IL-6* and *IL-1 β*) and chemokines (*CXCL1*, *MCPI* and *MIP2*) in the wound tissue. The concurrent reduction in macrophage (F4/80) and neutrophil (Ly6G) infiltration suggested that the sustained release of rhFGF1 effectively facilitated the transition from the inflammatory phase to the proliferative phase. This anti-inflammatory effect is consistent with the known role of FGF1 in metabolic regulation and immune modulation [11], but our data suggest that the delivery system amplifies this effect by maintaining local therapeutic concentrations. Furthermore, our results highlight the pivotal role of PLGA-rhFGF1 in promoting angiogenesis and re-epithelialization. The upregulation of angiogenesis-related factors (*VEGF*, *Ang-1*, and *eNOS*) and the increased density of CD31 vessels indicate robust neovascularization. Interestingly, Ang-1 and Ang-2 have been reported to maintain vascular maturation and induce vascular sprouting, respectively [39]. The observed increase in Ang-1 but not Ang-2 in our study suggests that PLGA-rhFGF1 not only induces new vessel formation but also promotes the maturation of a stable vascular network, which is crucial for

functional tissue recovery. Re-epithelialization is associated with epidermal proliferation and differentiation [23, 24]. In this study, the increased expression of keratin 10 and 14 confirms that the treatment effectively stimulates keratinocyte proliferation and differentiation, restoring the epidermal barrier.

In conclusion, our study demonstrates that the PLGA core-shell nanofiber system overcomes the instability limitations of rhFGF1. By providing a sustained release, PLGA-rhFGF1 synergistically modulates inflammation, enhances granulation tissue formation, and promotes re-epithelialization and angiogenesis. This system represents a promising, cost-effective therapeutic strategy for the treatment of chronic diabetic wounds.

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Author contributions

Yujie Zhang: Writing—original draft, Methodology, Conceptualization. Xianshi Wang: Methodology, Software. Shasha Ye: Methodology, Software. Jiana Li: Methodology. Junyi Wu: Methodology. Xiaoyan Bao: Writing—review & editing, Supervision. Yeli Zhao: Writing—review & editing, Visualization, Supervision, Funding acquisition.

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Data availability

All raw data related to this study can be obtained by contacting the corresponding author.

Declarations

Ethical approval

The animal study protocol was approved by the Animal Research Ethics Committee of Wenzhou Medical University (protocol code xmsq 2023-0390).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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